Security and Privacy for Machine Learning

Internal Assessment 1 — Problem Formulation and Preliminary Implementation

PrivBench-Adult

A Unified Privacy–Utility Evaluation of Differential, Local, Syntactic and Compressive Privacy for Tabular Machine Learning

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Executive Summary

Organisations increasingly train machine-learning (ML) models on personal records, and a growing set of privacy techniques claims to make this safe: differential privacy (DP) at training time, local DP at collection time, k-anonymity at publication time and compressive (lossy-projection) privacy at feature-release time. These techniques act at **different stages of the data pipeline, offer different kinds of guarantee and are usually evaluated in isolation with incomparable metrics**. A practitioner therefore cannot answer a simple question: *for a given utility target, which mechanism, at which stage, actually reduces which privacy risk?*

This project builds **PrivBench-Adult**, a unified evaluation framework in which one representative mechanism from each of Modules 2–5 is plugged into the same data pipeline, trained with the same model and attacked with the same Module-1 attack suite (membership inference, attribute inference and re-identification). The study uses the public UCI Adult census dataset (45,222 records after cleaning).

IA-1 status. The problem, objectives, scope and five-module mapping are fixed; the dataset is acquired, hashed, cleaned and split; all five modules have working code with 30 passing unit tests; and preliminary experiments have produced the first baseline evidence:

- Non-private baselines reach **84.9%** test accuracy (MLP). A fully grown random forest leaks membership (MIA advantage **0.172**), and leakage grows in a small-data regime (MLP: 0.159).
- **26.5%** of records are unique on seven quasi-identifiers, and the sensitive attribute *sex* can be inferred with **83.1%** accuracy even after the column is deleted, owing to proxies such as *relationship*.
- A from-scratch DP-SGD with a Rényi-DP accountant (validated against Opacus to 4 decimal places) keeps **85.0%** accuracy at ε = 1.
- At equal ε = 1, local-DP histograms are **280×** less accurate than central-DP histograms: the price of removing the trusted curator.

Rubric compliance map

Table 1. Mapping of IA-1 assessment stages to sections of this report.

IA-1 stage	Where addressed	Evidence
1. Problem definition & module mapping	Section 1	Problem statement, objectives, scope, outcomes, Table 2 (module mapping)
2. Conceptual understanding	Section 2	Concept notes, threat model, Figure 1 (workflow), treatment classification, justification
3. Dataset identification & preparation	Section 3	Source/licence table, SHA-256 hashes, preparation log, schema, limitations
4. Initial implementation	Section 4	Code structure, tools, design choices, execution instructions, run log, unit tests
5. Preliminary evaluation & documentation	Section 5	Baseline results, observations, limitations, IA-2 evaluation plan, contribution statement

1. Problem Definition and Module Mapping

1.1 Background and motivation

Removing names from a dataset is not anonymisation. Sweeney showed that 87% of Americans are unique on {ZIP, birth date, sex} [11]; Narayanan and Shmatikov de-anonymised Netflix users from sparse ratings [22]; and trained models themselves leak whether a person was in the training set (membership inference [19, 20]). Data-protection law now makes such leakage a compliance issue, e.g. India's Digital Personal Data Protection Act 2023 [31] and the EU GDPR [32]. The research community answered with a family of techniques, but each is typically presented with its own threat model and metric: ε for DP, k for anonymisation, reconstruction error for compressive privacy. They are rarely compared against the *same* attacker on the *same* data.

1.2 Problem statement

Problem statement. Given a tabular dataset of personal records used to train a binary classifier (income > 50K), determine — under one common data pipeline, one utility model and one attack suite — how much privacy protection (against membership inference, sensitive-attribute inference and re-identification) and how much utility loss each of four privacy-preserving mechanisms produces when applied at its natural pipeline stage: **local DP at collection**, **k-anonymity at publication**, **DP-SGD at training** and **compressive projection at feature release**; and identify whether a combination of mechanisms gives a better privacy–utility trade-off than any single one.

1.3 Significance

- **Practical decision support.** Engineers must choose *where* in a pipeline to spend a privacy budget. A side-by-side, attack-based comparison is directly actionable.
- **Exposes mismatched guarantees.** Syntactic methods (k-anonymity) protect identity but not attributes; DP protects membership but not population-level inference. A unified attack suite makes such gaps measurable rather than rhetorical.
- **Reproducible teaching artefact.** All five course modules become runnable, tested code on a real dataset, executable on a laptop CPU in minutes.

1.4 Objectives

Table 2. Project objectives and success criteria.

ID	Objective	Measurable success criterion
O1	Quantify the privacy exposure of the raw data and of non-private models	MIA AUC/advantage, QI uniqueness, attribute-inference accuracy reported with ≥3 seeds
O2	Implement one mechanism per module (M2–M5) from first principles	Unit-tested implementations; DP accountant matches a reference library
O3	Evaluate every mechanism with the same model and attack suite	One results table covering utility + all applicable attacks
O4	Characterise each privacy–utility trade-off across its parameter (ε, k, m)	Trade-off curves with mean ± sd across seeds
O5	Propose and test a combined mechanism	Hybrid configuration compared to the single mechanisms at matched ε

1.5 Scope

Table 3. Project scope.

In scope	Out of scope (with reason)
Tabular binary classification on UCI Adult	Images/text/LLMs — different threat surface; compute-heavy
Membership, attribute and re-identification attacks	Model inversion / reconstruction of training images (not meaningful for tabular one-hot data)
DP-SGD (central), GRR/OUE/PM (local), Mondrian k-anonymity + l-diversity, PCA/DCA projections	Cryptographic PPDM (secure multi-party computation, homomorphic encryption) — discussed analytically only
Laptop-CPU reproducibility; NumPy/scikit-learn only	Federated learning deployment, GPU-scale models

1.6 Expected outcomes

- A tested, open, reproducible Python framework (`src/privml`) covering all five modules.
- A provenance-documented, hashed dataset and preprocessing pipeline.
- Privacy–utility curves and a unified comparison table across mechanisms.
- Evidence-based guidance on which mechanism to apply at which pipeline stage.
- A research-manuscript draft (IA-2) describing the framework and the proposed hybrid.

1.7 Module-wise concept mapping

Table 4. Mapping of the problem to all five course modules (IA-1 version; refined in IA-2).

Module	Concepts used	Role in this project	Treatment
M1 Privacy considerations in ML	Threat models; membership inference; attribute inference; re-identification via quasi-identifiers; proxy attributes; privacy metrics	Defines *what* is being protected and supplies the attack suite used to measure leakage of every mechanism	Implemented + evaluated
M2 Differential privacy for ML	ε/δ-DP, sensitivity, Laplace & Gaussian mechanisms, composition, DP-SGD, Rényi-DP accounting	Protects the *training* stage; provides a formal, worst-case guarantee	Implemented + evaluated
M3 Local differential privacy	Randomised response, GRR, OUE, Piecewise mechanism, unbiased frequency estimation	Protects the *collection* stage when no trusted curator exists	Implemented + evaluated
M4 Privacy-preserving data mining	Quasi-identifiers, k-anonymity (Mondrian), l-diversity, t-closeness, information loss	Protects the *publishing* stage (release of a sanitised microdata table)	Implemented + evaluated
M5 Compressive privacy	Lossy dimensionality reduction (PCA), utility-supervised DCA, privacy-aware projection	Protects the *feature-release* stage: only a low-dimensional projection leaves the owner	Implemented (PCA) + planned (DCA variants)

2. Conceptual Understanding

2.1 Threat model

The data owner holds records x = (age, workclass, education-num, marital-status, occupation, relationship, race, sex, capital-gain, capital-loss, hours-per-week, native-country) with label y = 1[income > 50K]. The adversary follows **Kerckhoffs's principle**: they know the mechanism and its parameters, and they hold a small auxiliary sample from the same distribution (modelled by the 10% *public split*). Three adversary goals are considered:

Table 5. Threat model: adversaries evaluated in this project.

Attack	Adversary observes	Goal	Success metric
Membership inference (MIA)	Trained model's output on a target record	Decide whether the target was in the training set	ROC-AUC, advantage = max(TPR−FPR), TPR at 1% FPR
Attribute inference	Released record / representation (sex column possibly removed)	Infer the target's *sex*	Attacker accuracy and balanced accuracy vs. prior
Re-identification (linkage)	Released table + target's quasi-identifiers from an external source	Single out the target's record	Unique fraction, average prosecutor risk 1/|EC|

2.2 Concept notes

Module 1 — Privacy considerations in ML

Principle. Privacy risk in ML arises at every stage: collection, storage/publication, training and deployment. Over-fitted models assign lower loss to training points, which the loss-threshold MIA of Yeom et al. [20] exploits: predict *member* if $\ell(f(x), y) < \tau$. *Assumptions:* the attacker can query the model and knows the true label. *Advantages:* empirical attacks measure leakage in the units that matter to an individual. *Limitations:* an attack only gives a lower bound on leakage; a weak attack does not prove privacy.

Module 2 — Differential privacy

A randomised mechanism M is (ε, δ)-DP if for all neighbouring datasets D, D′ (differing in one record) and all output sets S:

$$Pr[M(D) \in S] \leq e^{\varepsilon} \cdot Pr[M(D') \in S] + \delta$$

The Laplace mechanism adds $\mathrm{Lap}(\Delta_1/\varepsilon)$ noise to a query of L1-sensitivity Δ_1. **DP-SGD** [2] trains a model privately: each step Poisson-samples records with rate q, clips every per-example gradient to norm C, sums, adds $N(0, \sigma^2 C^2 I)$ and updates. Privacy over T steps is tracked with Rényi DP [3, 4]; for integer order α the per-step RDP of the sub-sampled Gaussian is

$$\varepsilon_RDP(\alpha) = 1/(\alpha-1) \cdot \log \Sigma_{k=0}{}^{\alpha}\, C(\alpha,k)\, (1-q)^{(\alpha-k)}\, q^{k} \exp((k^2-k)/(2\sigma^2))$$

which composes additively over T steps and is converted to (ε, δ)-DP [5]. *Assumptions:* a trusted curator holds the raw data. *Advantages:* a formal guarantee that holds against any attacker, including future ones; composition. *Limitations:* utility cost at small ε, and disparate impact on minority groups [28]; the guarantee says nothing about population-level inferences.

Module 3 — Local differential privacy

Each user perturbs their own value before sending it, so even the collector is not trusted. Generalised randomised response (GRR) [6, 7] reports the true value with probability $p = e^{\varepsilon}/(e^{\varepsilon} + k - 1)$ and any other value otherwise; an unbiased frequency estimate is $\hat{f}_v = (\hat{c}_v/n - q)/(p - q)$. Optimised unary encoding (OUE) [8] is better for large domains ($k > 3e^{\varepsilon} + 2$). Numeric values use the Piecewise mechanism [9]. *Advantages:* no trusted party; deployed at scale (e.g. RAPPOR [10]). *Limitations:* estimation error grows as $O(\sqrt{n})$ rather than O(1), and multi-attribute records must split the budget across attributes.

Module 4 — Privacy-preserving data mining

A table is **k-anonymous** if every record is indistinguishable from at least k−1 others on its quasi-identifiers (QIs) [11]. Mondrian [12] achieves this by recursive median partitioning of the QI space. k-anonymity does not stop *homogeneity attacks* (all records in a class share the sensitive value), which motivates **l-diversity** [13] and **t-closeness** [14]. Information loss is measured with the Normalised Certainty Penalty (NCP) [15]. *Advantages:* the output is truthful microdata, interpretable by analysts. *Limitations:* no formal guarantee against background knowledge; it protects identity, not attributes.

Module 5 — Compressive privacy

Compressive privacy [17, 18] releases a lossy projection $Z = W^T(x - \mu)$ with $m \ll d$ dimensions instead of the record itself. **PCA** keeps directions of maximal variance (utility-agnostic); **Discriminant Component Analysis (DCA)** keeps directions maximising the utility-class discriminant ratio, i.e. the top generalised eigenvectors of $(S_B + \rho' I)w = \lambda(S_W + \rho I)w$. *Advantages:* data minimisation at source; cheap; composable with DP. *Limitations:* the guarantee is empirical (no ε), and a linear projection cannot remove non-linear information.

2.3 Methodology and workflow

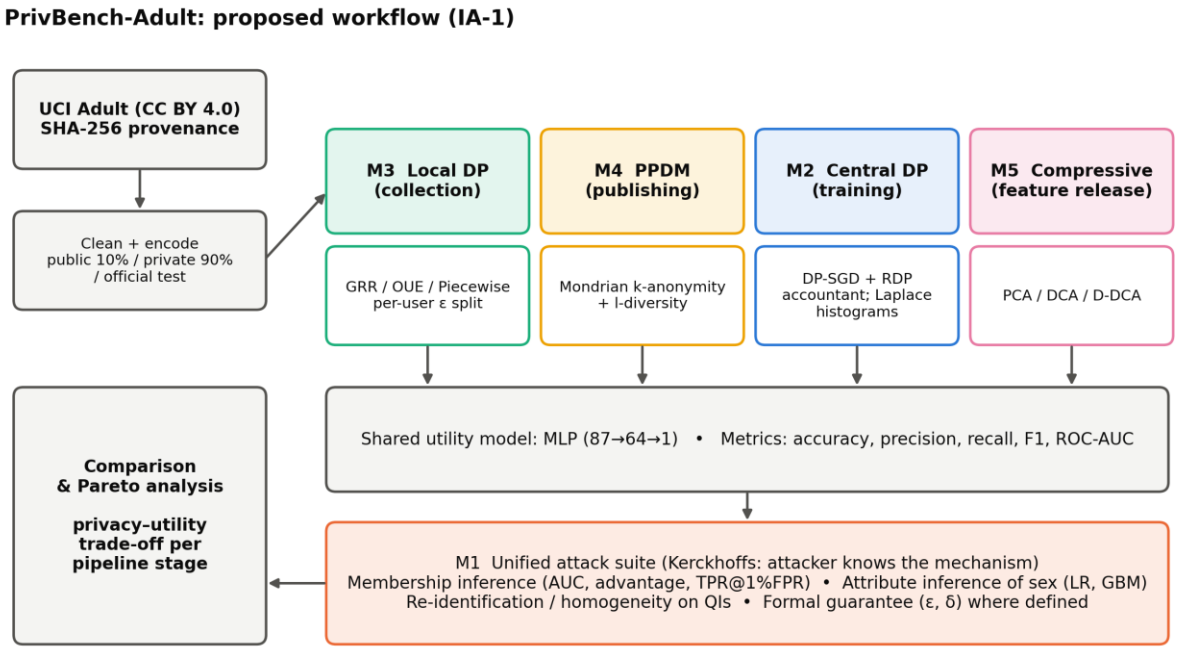


Figure 1. Proposed workflow. Each module protects a different pipeline stage; all feed the same utility model and the same Module-1 attack suite, which makes results directly comparable.

- **Step 1 — Data.** Download, hash, clean and split UCI Adult into public (10%), private-train (90%) and official test sets. Fit the encoder on the public split only.
- **Step 2 — Baseline exposure.** Measure QI uniqueness and attribute inference on raw data; train non-private LR/RF/MLP and attack them.
- **Step 3 — Mechanisms.** Apply each module's mechanism at its stage, sweeping its privacy parameter.
- **Step 4 — Common evaluation.** Train the same MLP on each protected output; report utility (accuracy, precision, recall, F1, AUC) and all applicable attacks.
- **Step 5 — Synthesis.** Compare trade-off curves and propose a combined mechanism.

2.4 Justification for concept selection

Table 6. Justification for the concepts and techniques selected.

Choice	Alternatives considered	Why this choice
DP-SGD + RDP accountant	Output perturbation, objective perturbation, PATE	Model-agnostic, standard in practice, tight accounting; PATE needs many teachers and unlabelled public data
GRR, OUE, Piecewise (LDP)	Basic RR, RAPPOR, Duchi et al.	GRR/OUE are the optimal frequency oracles for small/large domains [8]; PM is unbiased with bounded output [9]
Mondrian k-anonymity + l-diversity	Datafly, Incognito, t-closeness enforcement	Multidimensional partitioning gives far lower information loss than single-dimension generalisation; l-diversity directly targets homogeneity
PCA / DCA projections	Autoencoders, adversarial representation learning	Closed-form, deterministic, interpretable, and central to Kung's compressive-privacy framework [17]
Loss-threshold MIA	Shadow-model MIA [19], LiRA [21]	Strong for tabular models, assumption-light and cheap enough to run on every configuration

2.5 Treatment of concepts

Table 7. How each concept is treated (✓ = in IA-1; IA-2 = planned for IA-2).

Concept	Implemented	Demonstrated	Experimentally evaluated	Analytical
MIA, attribute inference, re-identification (M1)	✓		✓	
Laplace mechanism / DP histogram (M2)	✓	✓	✓	
DP-SGD + RDP accountant (M2)	✓		✓	
Gaussian mechanism (M2)	✓	✓		
GRR, OUE frequency oracles (M3)	✓		✓	
Piecewise mechanism, LDP record collection (M3)	✓		IA-2	
Mondrian k-anonymity, l-diversity (M4)	✓	✓	✓	
t-closeness (M4)	measured			✓
Secure multi-party computation / HE (M4)				✓
PCA (M5)	✓		✓	
DCA and privacy-aware projection (M5)	✓		IA-2	

3. Dataset Identification and Preliminary Preparation

3.1 Dataset and suitability

UCI Adult (Census Income) [23, 24] contains demographic and employment records extracted from the 1994 US Current Population Survey. It is suitable because (i) it contains genuine personal attributes, including classic quasi-identifiers (age, sex, race, marital status, native country) and a sensitive attribute (sex), which enables all three attack types; (ii) it is the most common benchmark in the DP, LDP and k-anonymity literature, so results can be compared with published work; (iii) it is openly licensed; and (iv)

at ~45k records it is large enough for DP-SGD to be meaningful and small enough for laptop experiments.

3.2 Source details

Table 8. Dataset source and provenance record (machine-readable copy: data/provenance.json).

Field	Value
Name	Adult (Census Income)
Landing page	https://archive.ics.uci.edu/dataset/2/adult
Download URL	https://archive.ics.uci.edu/static/public/2/adult.zip
DOI	10.24432/C5XW20
Creators	Barry Becker, Ronny Kohavi (extracted from the 1994 US Census database)
Publisher	UCI Machine Learning Repository, University of California, Irvine
Licence	Creative Commons Attribution 4.0 International (CC BY 4.0)
Original collection	1994 Current Population Survey (US Census Bureau); extraction by Becker (1996) with filters AGI>100, AFNLWGT>1, AHRSWK>0
Version	UCI static release id 2 (donated 1996-04-30)
Retrieved on / method	2026-10-20 — HTTPS download of the official UCI archive by scripts/01_prepare_data.py (urllib)
SHA-256 (adult.zip)	`7537312dd56c2b98035880805ce99e68183a30ee468aa5329d6df0fbb3cc21bb`

3.3 Schema and attribute roles

Table 9. Attribute schema and roles.

Attribute	Type	Role in this study
age	numeric (17–90)	feature; quasi-identifier
workclass	categorical (7)	feature; quasi-identifier
education-num	numeric (1–16)	feature; quasi-identifier
marital-status	categorical (7)	feature; quasi-identifier
occupation	categorical (14)	feature
relationship	categorical (6)	feature; strong proxy of sex
race	categorical (5)	feature; quasi-identifier
sex	binary	feature; quasi-identifier; **sensitive attribute** for attribute inference
capital-gain / capital-loss	numeric	feature
hours-per-week	numeric (1–99)	feature
native-country	categorical (41)	feature; quasi-identifier
income	binary	**utility label** (>50K = 1); sensitive value for l-diversity
fnlwgt, education	—	dropped (census sampling weight; duplicate of education-num)

3.4 Preliminary preparation log

All steps are executed by `scripts/01_prepare_data.py`; the log below is copied verbatim from its output and stored in `data/provenance.json`.

Table 10. Data preparation log.

#	Step
1	Loaded raw files: train=32561 rows, test=16281 rows, 15 columns
2	train: 2399 rows with missing values ('?') in {'workclass': 1836, 'occupation': 1843, 'native-country': 583}
3	test: 1221 rows with missing values ('?') in {'workclass': 963, 'occupation': 966, 'native-country': 274}
4	Dropped incomplete rows: train 32561->30162, test 16281->15060
5	Exact duplicate rows in train: 23 (kept: duplicates are distinct census respondents with identical attributes)
6	Dropped columns ['fnlwgt', 'education']; binarised target income>50K -> 1
7	Positive rate: train 0.2489, test 0.2457
8	Split official train into public=3016 / private=27146 (seed 0); official test kept as test=15060
9	Encoded feature dimension: 87 (5 standardised numeric + one-hot categoricals)

Table 11. Dataset size before and after preparation.

Quantity	Original	After preparation
Training rows	32,561	30,162 (public 3,016 + private 27,146)
Test rows	16,281	15,060
Attributes	14 + label	12 features + label (87 after one-hot encoding)
Positive rate (>50K)	—	train 0.2489, test 0.2457
Male share	—	0.6757

Why a public split? 10% of the official training file is set aside as *public* data. The encoder's means/standard deviations and the compressive projections are fitted on it, so no statistic of the private training data leaks through preprocessing (a common, often-ignored leak in DP pipelines). The attacker's auxiliary data is drawn from the same split.

3.5 Initial exposure analysis

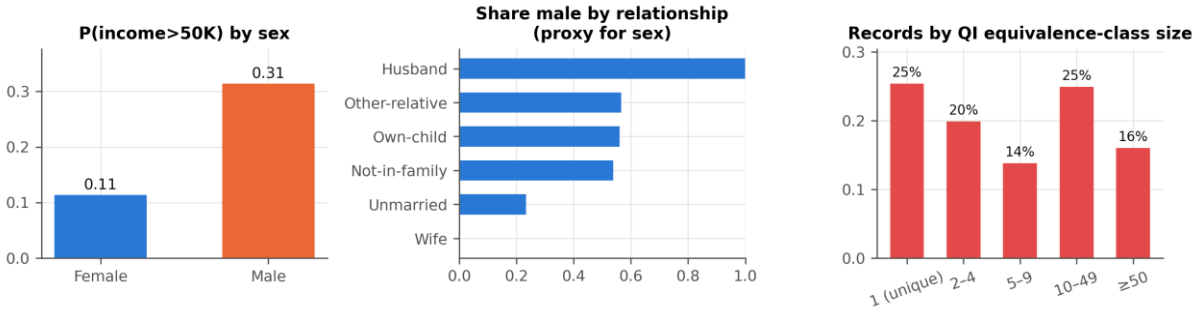


Figure 2. Left: income rate differs strongly by sex. Centre: relationship is a near-deterministic proxy of sex (Husband → male, Wife → female). Right: a quarter of records are unique on seven QIs.

3.6 Known dataset limitations

- Collected in 1994 in the US; distributions are dated and not representative of other populations [25].
- Records with missing values (7.4% of rows) were dropped; this may bias the sample towards respondents with complete employment information.
- The dataset is already public, so the experiments *simulate* a private dataset; absolute risk numbers describe the methodology, not harm to real respondents.
- Sex is recorded as binary; this reflects the source and limits the attribute-inference analysis.

4. Initial Implementation

4.1 Tools and libraries

Table 12. Tools and libraries.

Tool	Version	Used for
Python	3.12.4	Language
NumPy	2.x	MLP, DP-SGD, LDP mechanisms, projections
SciPy	1.18	log-sum-exp & log-gamma for the accountant; generalised eigensolver
pandas	2.2	Data handling, Mondrian tables, result CSVs
scikit-learn	1.8	LR / RF baselines, gradient-boosting attacker, metrics
matplotlib	3.9	Figures
pytest	9.0	Unit tests
Opacus	1.5.2	*Only* as an external reference to validate the accountant (not a runtime dependency)

4.2 Code structure

Table 13. Repository structure.

Path	Purpose
`scripts/01_prepare_data.py`	Download UCI archive, SHA-256 hashing, cleaning, provenance record
`scripts/02_run_ia1.py`	IA-1 preliminary experiments P1–P6
`scripts/03_run_ia2.py`	IA-2 full experiments E1–E8 (`--quick` for smoke test)
`scripts/04_make_figures.py`	All figures from the result CSVs
`scripts/05_build_reports.py`	Regenerates these Word reports from results
`src/privml/data.py`	Schema, cleaning, public/private/test splits, encoder fitted on public split
`src/privml/attacks.py`	M1: membership inference, attribute inference, re-identification risk
`src/privml/dp.py`	M2: Laplace/Gaussian mechanisms, DP histogram, RDP accountant, DP-SGD
`src/privml/ldp.py`	M3: GRR, OUE, Piecewise mechanism, LDP record perturbation
`src/privml/ppdm.py`	M4: Mondrian k-anonymity, l-diversity, NCP, homogeneity, t-closeness
`src/privml/compressive.py`	M5: PCA, DCA, Desensitised DCA (D-DCA), reconstruction error
`src/privml/models.py`	NumPy MLP with closed-form per-example gradients; LR/RF baselines
`src/privml/experiments.py`	Experiment definitions shared by IA-1 and IA-2 runners
`tests/`	30 unit tests (accountant vs Opacus, estimator unbiasedness, k-anonymity, gradients)

4.3 Key implementation choices

- **DP-SGD without a deep-learning framework.** For a one-hidden-layer MLP the per-example gradient norm has a closed form: $\|\nabla \ell_i\|^2 = \|x_i\|^2\|\delta_i\|^2 + \|\delta_i\|^2 + e_i^2\|h_i\|^2 + e_i^2$, where $e_i = p_i - y_i$, h_i is the hidden activation and $\delta_i = e_i w_2 \odot 1[a_i > 0]$. The clipped sum is $X^T(c \odot \Delta)$. This avoids materialising per-example weight matrices and runs a 20-epoch DP-SGD in ~1.5 s on a CPU. It also removed an incompatibility between the installed PyTorch build and NumPy 2.
- **Poisson sub-sampling** (each record included independently with probability q) is used so that the implementation matches the assumption of the RDP accountant exactly.
- **Accountant validated externally.** ε values for four (q, σ, T) settings match Opacus 1.5.2 to four decimal places (`tests/test_dp.py`).
- **Noise calibration by bisection.** `calibrate_sigma` finds the smallest σ meeting a target ε.

- **Set-valued generalisation** for categorical QIs in Mondrian (Adult has no published hierarchies), encoded as fractional multi-hot vectors so models can train on anonymised data.
- **Seeds everywhere.** Every random step takes an explicit seed; results are reported over seeds.

4.4 Inputs, processing and outputs

Table 14. Inputs, processing steps and outputs of each component.

Component	Input	Processing	Output
Data preparation	UCI archive (URL)	hash → parse → drop missing → drop 2 columns → split	clean CSVs, provenance.json
Encoder	public split	standardise 5 numeric, one-hot 7 categorical	87-dim feature matrix
Baselines	private-train features/labels	fit LR, RF, MLP	models
DP-SGD	features, labels, target ε, δ	calibrate σ, Poisson sample, clip, noise, SGD	private MLP + (ε, δ, σ, steps)
Histogram release	categorical column, ε	Laplace (central) or GRR/OUE perturb + debias (local)	estimated frequencies
Mondrian	raw table, k, (l)	recursive median / set splits on 7 QIs	equivalence classes, generalised table
PCA	public features, m	eigen-decomposition of covariance	projection W (87×m)
Attack suite	model or released data	loss-threshold MIA; GBM/LR attribute attacker; class-size risk	privacy metrics

4.5 Execution instructions

```
# 1. install dependencies (Python 3.10+)
pip install -r requirements.txt

# 2. download, hash and clean the dataset  (~10 s)
python scripts/01_prepare_data.py

# 3. run the IA-1 preliminary experiments  (~1 min)
python scripts/02_run_ia1.py

# 4. make figures and run the unit tests
python scripts/04_make_figures.py
python -m pytest
```

4.6 Output log (excerpt)

```
[    0.2s] Data: public=3016 private=27146 test=15060 d=87
[    0.2s] === P1 baselines ===
[   21.7s] --- P1 baselines done in 21.5s
[   21.7s] saved p1_baselines.csv (18 rows)
[   21.7s] === P2 raw-data exposure ===
[   23.0s] --- P2 raw-data exposure done in 1.3s
[   23.0s] saved p2_exposure.csv (5 rows)
[   23.0s] === P3 DP-SGD (preliminary) ===
[   48.9s] --- P3 DP-SGD (preliminary) done in 25.9s
[   48.9s] saved p3_dpsgd.csv (18 rows)
[   48.9s] === P4 central vs local histogram ===
[   50.4s] --- P4 central vs local histogram done in 1.5s
[   50.4s] saved p4_marginals.csv (420 rows)
[   50.4s] === P5 k-anonymity (preliminary) ===
[   54.9s] --- P5 k-anonymity (preliminary) done in 4.5s
[   54.9s] saved p5_kanon.csv (2 rows)
[   54.9s] === P6 PCA compression (preliminary) ===
[   56.4s] --- P6 PCA compression (preliminary) done in 1.5s
[   56.4s] saved p6_pca.csv (3 rows)
[   56.4s] IA-1 run complete
```

4.7 Verification

`python -m pytest` runs **30 unit tests, all passing**. They check: the accountant against Opacus; ε monotonicity and σ calibration; analytic per-example gradients against finite differences; Laplace noise scale; unbiasedness of the GRR, OUE and Piecewise estimators; that Mondrian output is a disjoint k-anonymous partition with max risk ≤ 1/k; that l-diversity removes homogeneous classes; that D-DCA removes a linear privacy signal on synthetic data; that the MIA detects a memorising model; and that dataset hashes match the provenance record.

5. Preliminary Evaluation and Documentation

5.1 Evaluation measures

Table 15. Evaluation measures.

Dimension	Metric	Interpretation
Utility	Accuracy, precision, recall, F1 (positive = >50K), ROC-AUC	F1 matters because only ~25% of records are positive
Membership privacy	MIA ROC-AUC; advantage = max(TPR−FPR); TPR at 1% FPR	0.5 / 0 / 0.01 = no leakage
Attribute privacy	Attacker accuracy and balanced accuracy on sex	0.5 balanced accuracy = no leakage
Identity privacy	Fraction unique on QIs; average & max prosecutor risk	max risk ≤ 1/k under k-anonymity
Disclosure (M4)	Homogeneity rate; t-closeness	share of records whose income is revealed
Information loss	NCP (M4); normalised reconstruction MSE (M5)	0 = no loss
Formal guarantee	(ε, δ) for DP/LDP mechanisms	smaller ε = stronger guarantee

5.2 Baseline results (non-private models)

Table 16. Non-private baselines (mean ± sd over 3 seeds). Small-data regime = random 1,000-record training subset with a high-capacity MLP (256 hidden units, 200 epochs).

Model	Accuracy	F1	AUC	MIA AUC	MIA adv.	Train loss	Test loss
LR (n=27,146)	0.847 ± 0.000	0.659	0.903	0.503 ± 0.009	0.014	0.326	0.331
RF (n=27,146)	0.841 ± 0.001	0.654	0.889	0.603 ± 0.008	0.172	0.106	0.404
MLP (n=27,146)	0.849 ± 0.002	0.674	0.907	0.508 ± 0.010	0.022	0.297	0.326
LR (n=1,000)	0.846 ± 0.000	0.659	0.895	0.502 ± 0.015	0.032	0.301	0.331
RF (n=1,000)	0.836 ± 0.003	0.641	0.875	0.640 ± 0.012	0.256	0.093	0.371
MLP (n=1,000)	0.822 ± 0.003	0.619	0.869	0.552 ± 0.015	0.159	0.076	0.631

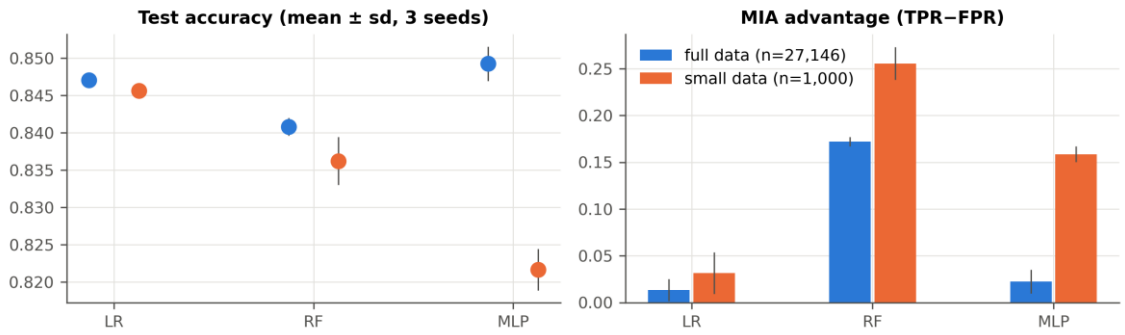


Figure 3. Baseline utility and membership-inference advantage.

With the full training set, LR and the MLP generalise well (train/test loss gap ≈ 0.03), and average-case MIA is near chance (AUC ≈ 0.508). The fully grown random forest memorises its training data (train loss 0.106 vs test 0.404) and leaks membership (advantage 0.172). In the small-data regime the MLP's advantage rises to 0.159 and the RF's to 0.256: **leakage is driven by the generalisation gap**, consistent with Yeom et al. [20].

5.3 Privacy exposure of the raw data

Table 17. Exposure of the unprotected data.

Analysis	Result
Records unique on 7 QIs	26.5%
Average prosecutor re-identification risk	0.381
Number of QI equivalence classes	10,340
Infer sex — all features, linear attacker	100.0%
Infer sex — sex column dropped, linear attacker	84.3%
Infer sex — sex column dropped, GBM attacker	83.1%
Majority-class prior for sex	67.4%

Two Module-1 lessons follow immediately: (i) removing direct identifiers does not anonymise — one in four records can be singled out by linkage; (ii) deleting a sensitive column ("fairness/privacy through unawareness") is weak protection, because proxies retain most of the signal (83.8% balanced accuracy versus 50% for no information).

5.4 Preliminary DP-SGD results

Table 18. DP-SGD on the full training set (δ = 1e-5, 20 epochs, batch 256, C = 1; mean of 3 seeds).

Target ε	σ	Accounted ε	Accuracy	F1	MIA AUC	MIA adv.
0.5	3.446	0.500	0.848	0.659	0.503	0.016
1	1.931	1.000	0.850	0.665	0.503	0.014
2	1.193	2.000	0.851	0.665	0.503	0.014
4	0.852	3.995	0.851	0.666	0.503	0.014
8	0.665	7.999	0.851	0.665	0.503	0.015
∞ (clip only)	0	∞	0.851	0.665	0.503	0.015

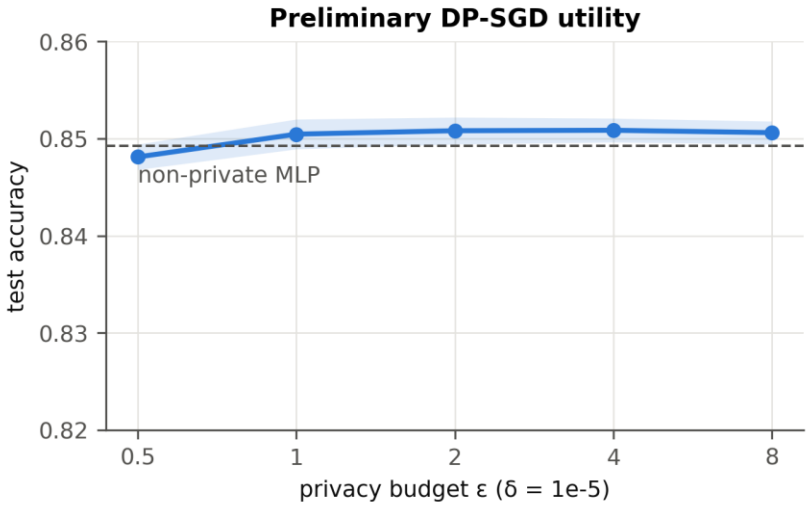


Figure 4. Preliminary DP-SGD utility versus privacy budget.

Worked example of the accounting: with n = 27,146, batch 256 ⇒ q = 0.00943, T = round(20/q) = 2,121 steps. For ε = 1 the bisection returns σ = 1.931; the accountant gives ε = 1.0000 at δ = 10^{-5}.

5.5 Central versus local DP for histogram release

Table 19. Mean L1 error of the 14-category occupation histogram (20 repetitions).

ε	Central Laplace	Local GRR	Local OUE	GRR / central
0.1	0.00551	2.2494	1.4546	408×
0.25	0.00221	0.8390	0.5773	380×
0.5	0.00110	0.3788	0.3105	343×
1	0.00055	0.1544	0.1469	280×
2	0.00028	0.0535	0.0608	194×
4	0.00014	0.0132	0.0261	96×
8	0.00007	0.0017	0.0160	25×

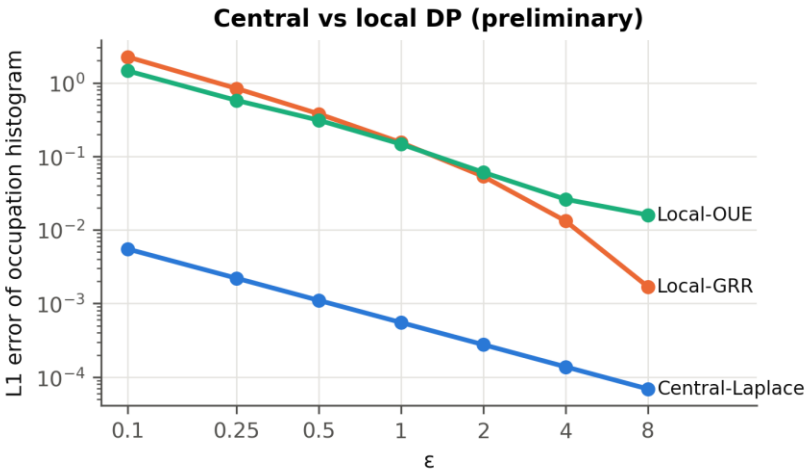


Figure 5. Central vs. local DP error (log–log).

Central DP is two to three orders of magnitude more accurate at the same ε. Between the local oracles, OUE wins for ε ≤ 1 and GRR for ε ≥ 2, matching the theoretical cross-over k = 3e^ε + 2 (for k = 14 the cross-over is ε ≈ 1.3) [8].

5.6 Preliminary k-anonymity

Table 20. Mondrian k-anonymity on the 7 QIs (private training split).

k	Classes	NCP	Homogeneity rate	Unique fraction	Avg risk	Max risk
5	2663	0.018	39.6%	0.0%	0.098	0.200
10	1474	0.032	30.2%	0.0%	0.054	0.100

Table 21. First records of the 10-anonymous release (generalised QIs shown as ranges/sets).

age	workclass	education-num	marital-status	race	sex	native-country	income
32-37	Private	11-16	{Married-AF-spouse,Married-spouse-absent,Separated,Widowed}	White	Female	United-States	0
55	Private	9	Married-civ-spouse	White	Male	United-States	1
27-37	State-gov	11-13	Never-married	White	Female	United-States	0
43-45	Private	9	Never-married	White	Male	{Poland,United-States}	0
21-26	Self-emp-not-inc	9	Married-civ-spouse	White	Male	United-States	0
22-31	Federal-gov	11-13	Never-married	White	Male	{Japan,United-States}	0
52-54	Self-emp-not-inc	9	Married-civ-spouse	White	Male	United-States	0
52-54	Private	10	Divorced	White	Female	{Germany,Italy,United-States}	0

Mondrian removes all unique records with little information loss (NCP 0.032 at k = 10). However, **30.2% of records sit in classes where every member has the same income**, so their income is disclosed despite k-anonymity — the homogeneity attack that motivates l-diversity in IA-2.

5.7 Preliminary compressive privacy (PCA)

Table 22. PCA on all 87 features: sensitive-attribute leakage from the projection.

m	Reconstruction NMSE	Infer sex (LR) bal. acc.	Infer sex (GBM) bal. acc.
5	0.355	0.784	0.860
10	0.186	1.000	0.996
20	0.071	1.000	0.998

Utility-agnostic PCA is *not* privacy-preserving: at m ≥ 10 the sex column survives the projection almost intact (balanced accuracy ≈ 1.0). Compression alone only helps when m is very small, which also destroys utility. A projection that is explicitly *penalised* for retaining the sensitive attribute is required.

5.8 Preliminary observations

Table 23. Preliminary observations and their consequences.

#	Observation	Consequence for IA-2
O1	Average-case MIA on well-generalising models is near chance; leakage concentrates in high-capacity / small-data settings	Add a small-data regime to every DP experiment; report TPR at 1% FPR
O2	DP-SGD costs < 0.3 points of accuracy for ε ≥ 0.5 on Adult	Extend the ε grid down to 0.05 to expose the real trade-off
O3	k-anonymity removes uniqueness but leaves high homogeneity	Evaluate l-diversity; measure attribute inference on anonymised data
O4	PCA retains the sensitive attribute	Design a privacy-aware projection (Desensitised DCA) and a hybrid with DP
O5	Local DP loses ~100–1000× accuracy vs central DP in histograms	Measure the effect on model training with LDP-collected records

5.9 Known limitations at IA-1

- Only three seeds; confidence intervals are wide for small effects.
- Hyper-parameters of DP-SGD (lr = 0.05, C = 1, 20 epochs) were chosen by a brief sanity run; tuning on private data has its own privacy cost [27] and is not yet accounted for.
- The loss-threshold MIA is a lower bound on leakage; stronger attacks (LiRA [21]) may find more.

- LDP record collection, DCA variants and the hybrid have code but no experiments yet.

5.10 Proposed evaluation plan for IA-2

Table 24. Evaluation plan for IA-2.

Exp.	Question	Design	Metrics
E1	How much do non-private models leak?	LR / RF / MLP, full and small data, 5 seeds	utility, MIA
E2	Privacy–utility curve of DP-SGD	$\varepsilon \in \{0.05 \dots 8, \infty\}$, both regimes, 5 seeds	utility, MIA, σ
E3	Sensitivity to DP-SGD hyper-parameters	clip C, learning rate, epochs at $\varepsilon = 1$	accuracy
E4	Cost of removing the trusted curator	Laplace vs GRR vs OUE, 4 attributes, 50 reps	L1, MSE
E5	Can we train on LDP-collected records?	$\varepsilon_total \in \{1 \dots 128\}$, with/without label perturbation	utility, MIA, attribute inference
E6	k-anonymity and l-diversity trade-off	$k \in \{2 \dots 250\}$, $l \in \{\text{none}, 2\}$	NCP, risk, homogeneity, utility, attribute inference
E7	Can a projection hide sex but keep income?	PCA / DCA / D-DCA, $m \in \{2 \dots 40\}$, γ sweep	utility, LR & GBM attribute inference, reconstruction
E8	Does compression help DP-SGD?	D-DCA → DP-SGD vs DP-SGD across ε	utility, MIA, attribute inference

5.11 Individual contribution statement

This is an individual project. I defined the problem and threat model, selected the dataset and documented its provenance, and designed the unified evaluation methodology. I implemented the `src/privml` package — the data pipeline with a public split, the attack suite, the RDP accountant and DP-SGD (including the closed-form per-example gradients of the NumPy MLP), the GRR/OUE/Piecewise mechanisms, Mondrian k-anonymity with l-diversity, and the PCA/DCA projections — together with the unit tests, experiment scripts, figures and this report. Third-party libraries are used only for standard components (NumPy/SciPy linear algebra, scikit-learn baselines and attackers, matplotlib); Opacus was used solely as an external check of the accountant.

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